
Multidimensional design

Knowledge Objectives

1. Justify the usefulness of multidimensional analysis against operational databases and spreadsheets
2. Define OLAP (On-Line Analytical Processing)
3. Describe a data cube
4. Describe the most typical multidimensional operations over the cube
5. Distinguish the main kinds of multidimensional tools (ROLAP, MOLAP and HOLAP)

Understanding Objectives

1. Explain the meaning of a star (or its variants snowflake and galaxy) shape multidimensional schema
2. Translate a multidimensional UML diagram into a relational star-join schema
3. Elaborate on the advantages and disadvantages of ROLAP in front of MOLAP

Motivation and definition

Spreadsheet (I)

	A	B	C	D	E	F	G	H	...
1									
2		x							
3		y							
4		z							
5									
6									
...									

SUM(B2:B4)



SUM(B2; G3; F6)

Spreadsheets (II)

- Absence of metadata
 - Rows and columns without associated meaning
 - Difficult query and interpretations
- Limited amount of data
 - M\$Excel ($65,000 \times 256 = 16,000,000$ cells)
- The position limits operations
- Aggregation hierarchies are not managed

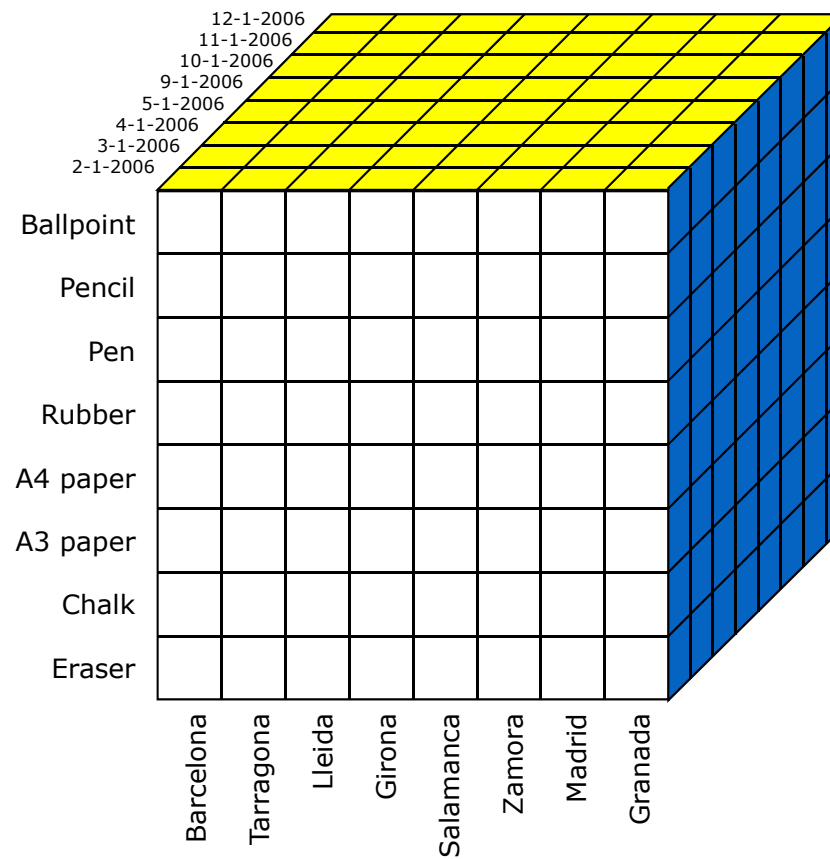
FASMI test

- Fast
- Analysis
- of Shared
- Multidimensional
- Information

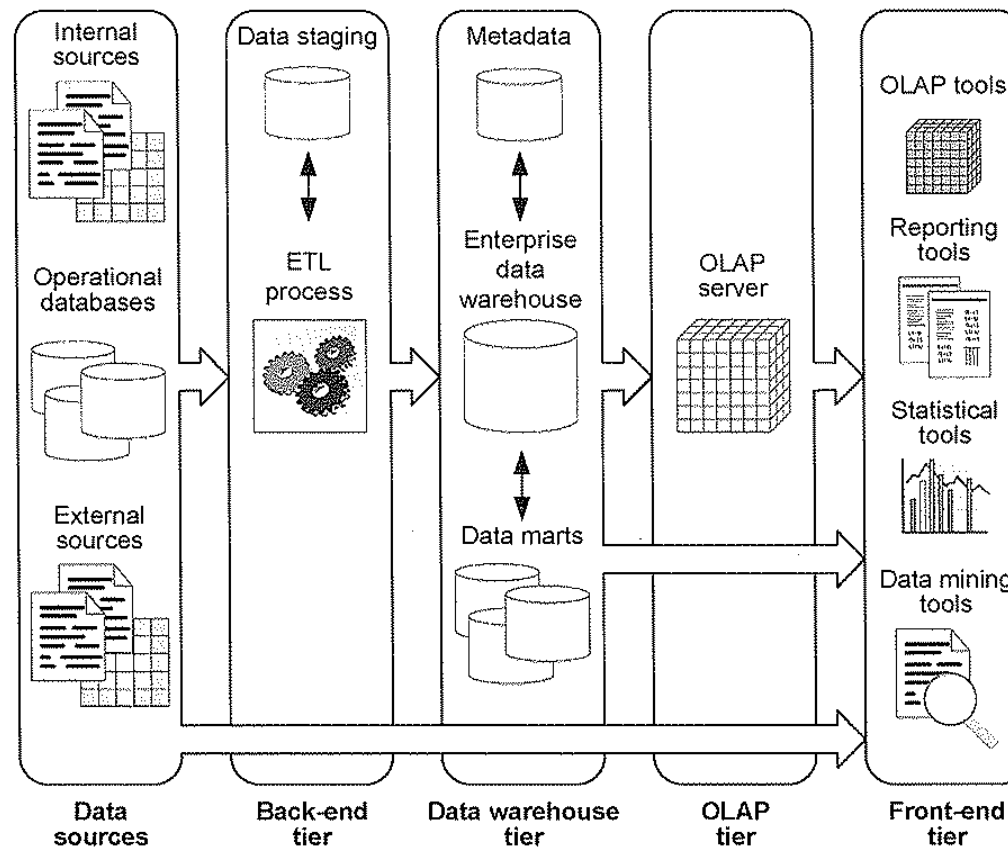
Nigel Pendse, 1995

Spreadsheet -> FAMSI
Operational DB -> SIFAM

Cube

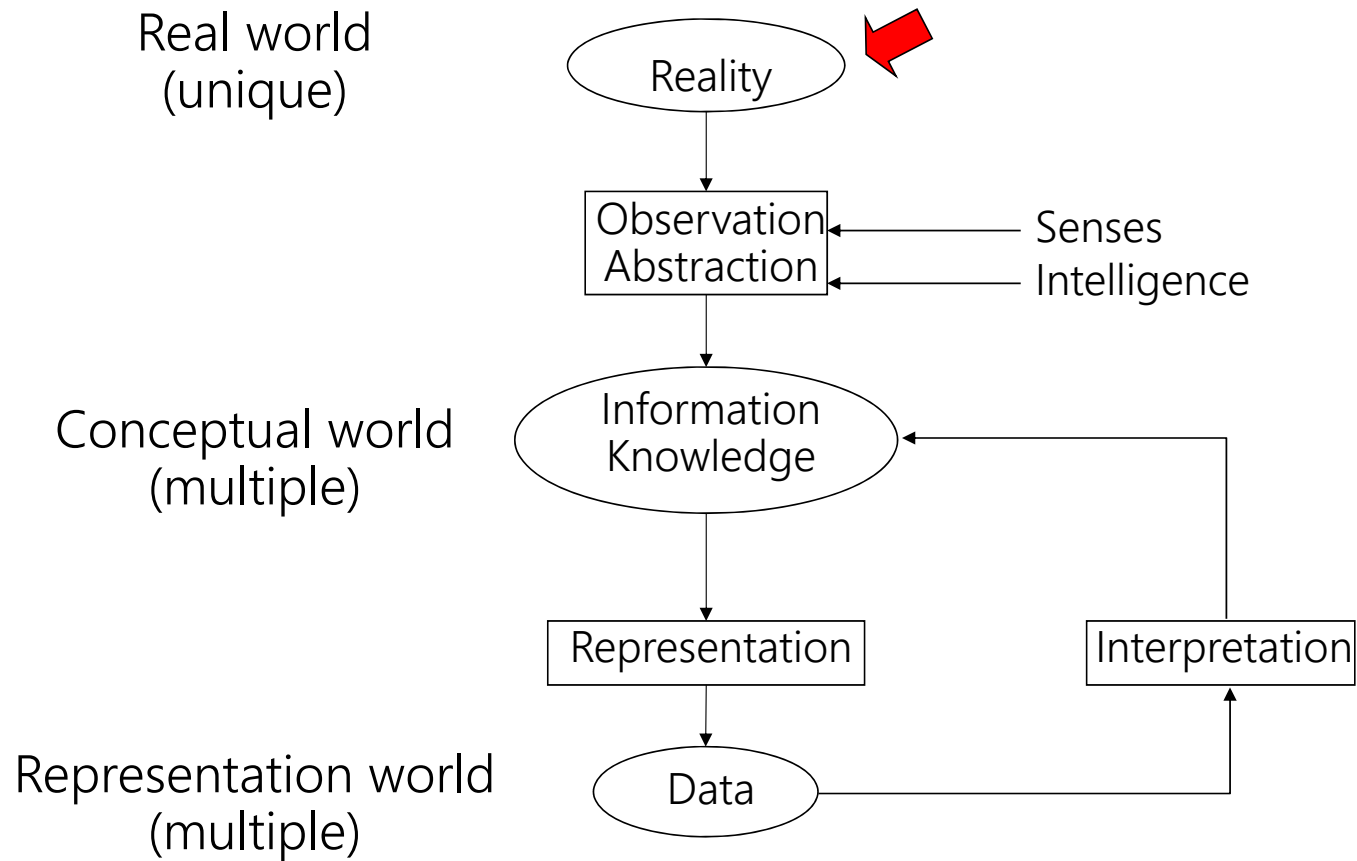


Reference architecture



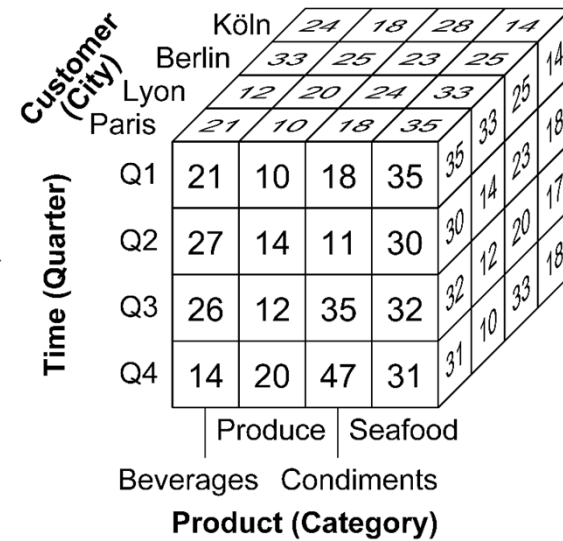
Real world

Three worlds

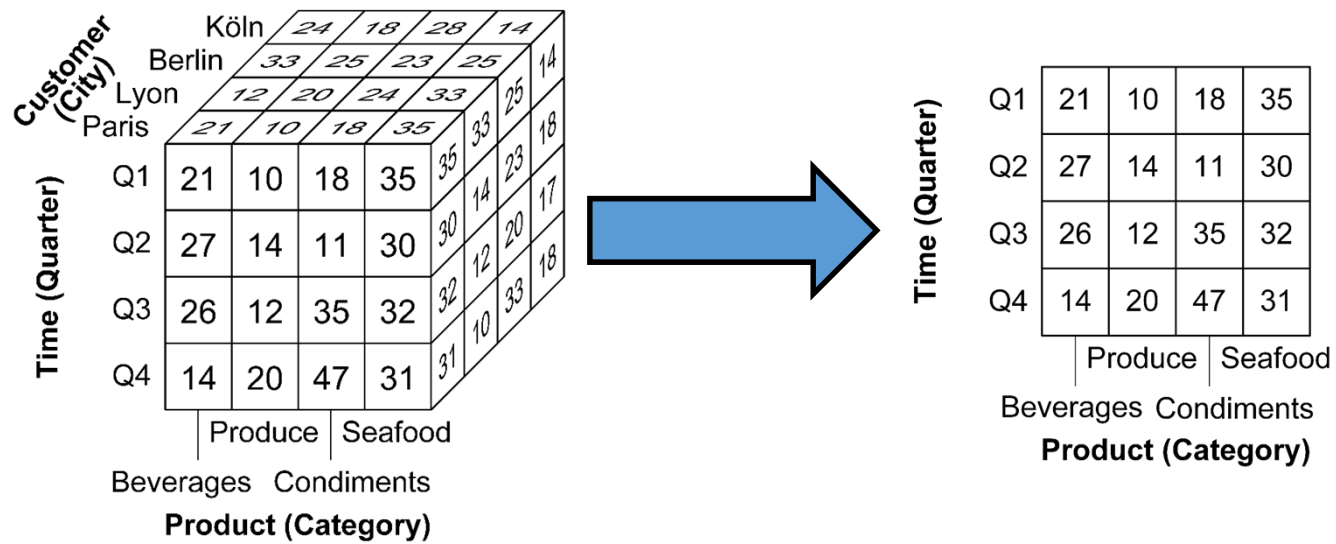


Cube view

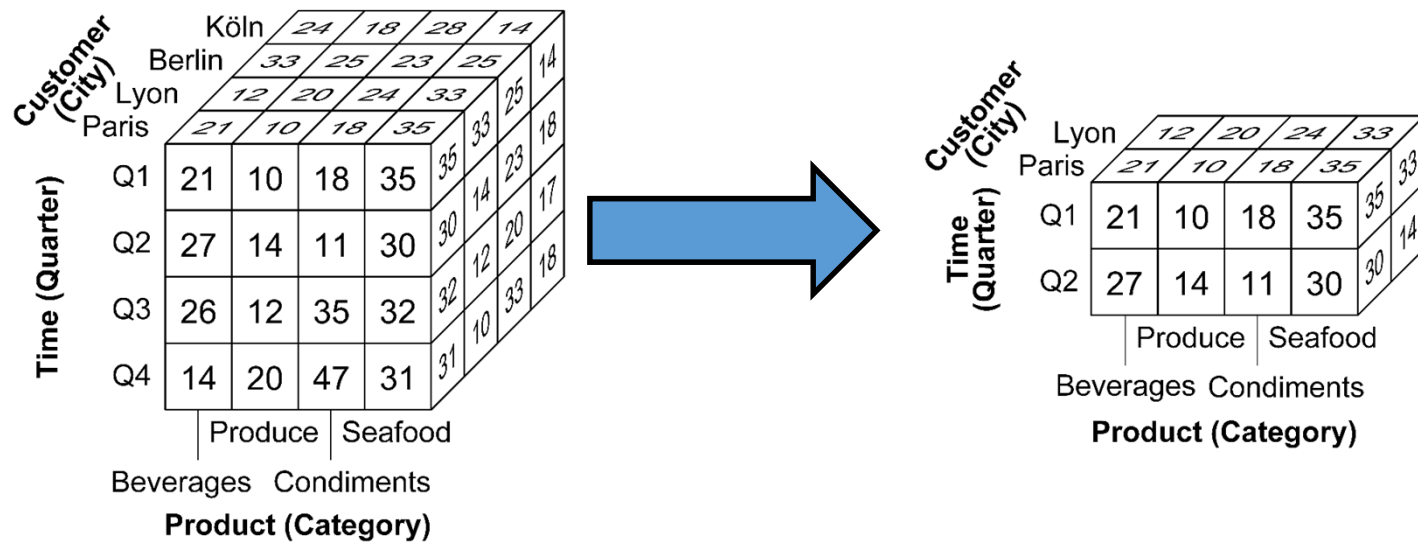
		Q1	Q2	Q3	Q4
Paris	Beverages	21	27	26	14
	Produce	10	14	12	20
	Condiments	18	11	35	47
	Seafood	35	30	32	31
Lyon	Beverages	12			
	Produce	20			
	Condiments	24			
	Seafood	33	14	12	10
Berlin	Beverages	33			
	Produce	25			
	Condiments	23			
	Seafood	25	23	20	33
Köln	Beverages	24			
	Produce	18			
	Condiments	28			
	Seafood	14	18	17	18



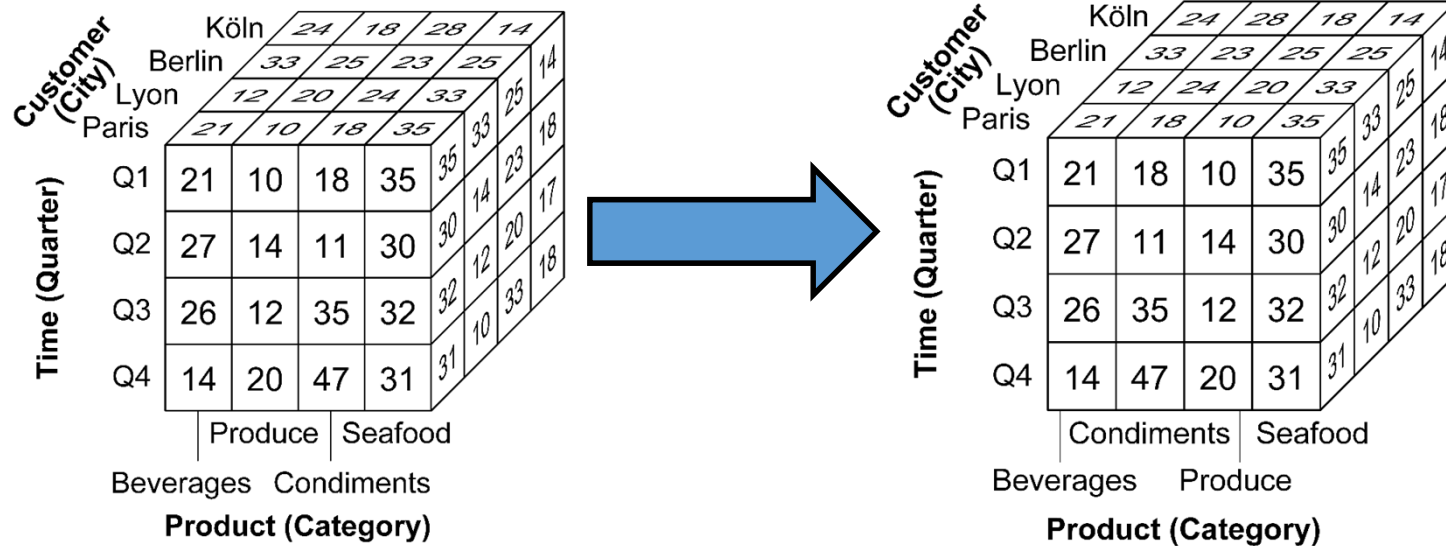
Slice



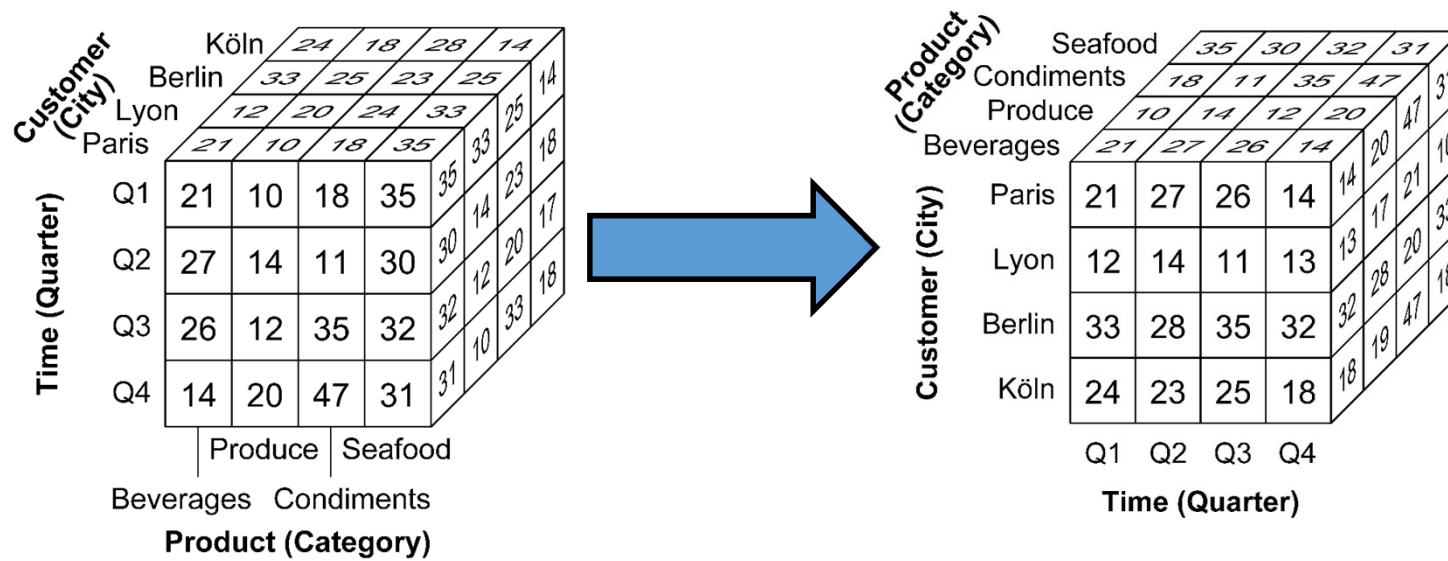
Dice



Sort



Pivot (I)



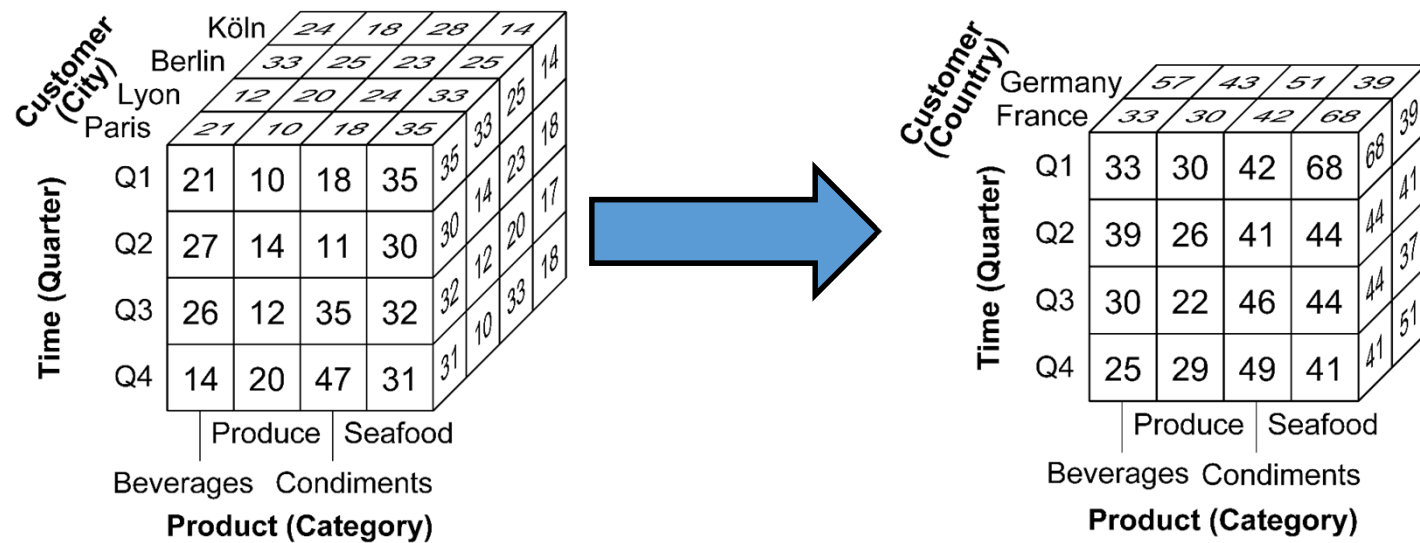
Pivot (II)

		Q1	Q2	Q3	Q4
Paris	Beverages	21	10	18	35
	Produce	27	14	11	30
	Condiments	26	12	35	32
	Seafood	14	20	47	31
Lyon	Beverages	12	20	24	33
	Produce				
	Condiments				
	Seafood				
Berlin	Beverages	33	25	23	25
	Produce				
	Condiments				
	Seafood				
Köln	Beverages	24	18	28	14
	Produce				
	Condiments				
	Seafood				

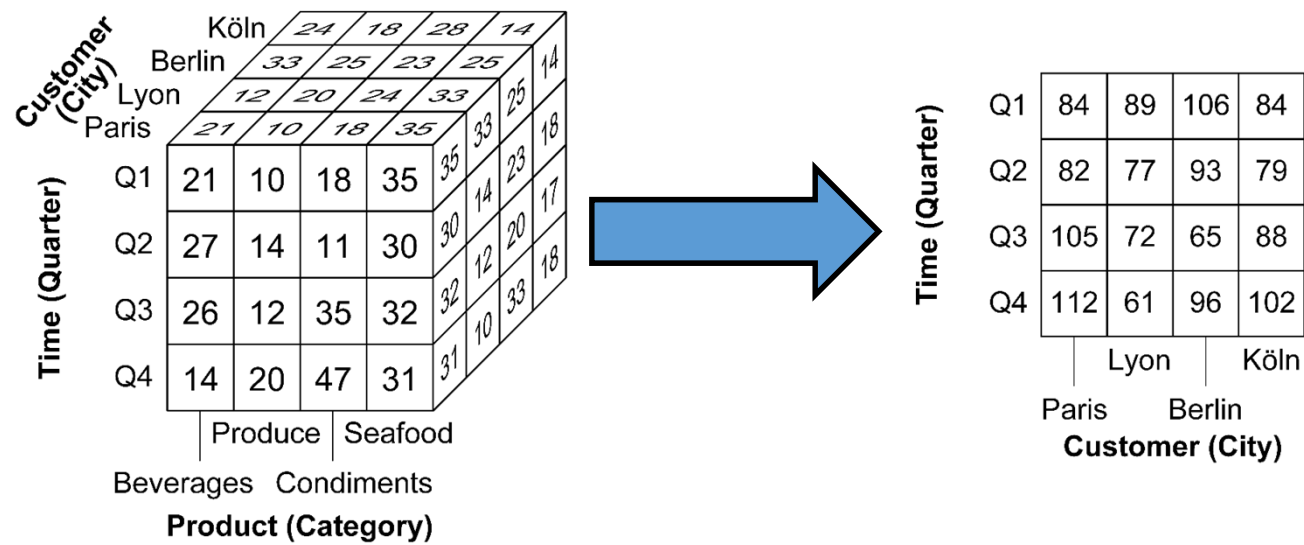


		Paris	Lyon	Berlin	Köln
Beverages	Q1	21	27	26	14
	Q2	12	14	11	12
	Q3	33	28	35	32
	Q4	24	23	25	18
Produce	Q1	10	14	12	20
	Q2				
	Q3				
	Q4				
Condiments	Q1	18	11	35	47
	Q2				
	Q3				
	Q4				
Seafood	Q1	35	30	32	31
	Q2				
	Q3				
	Q4				

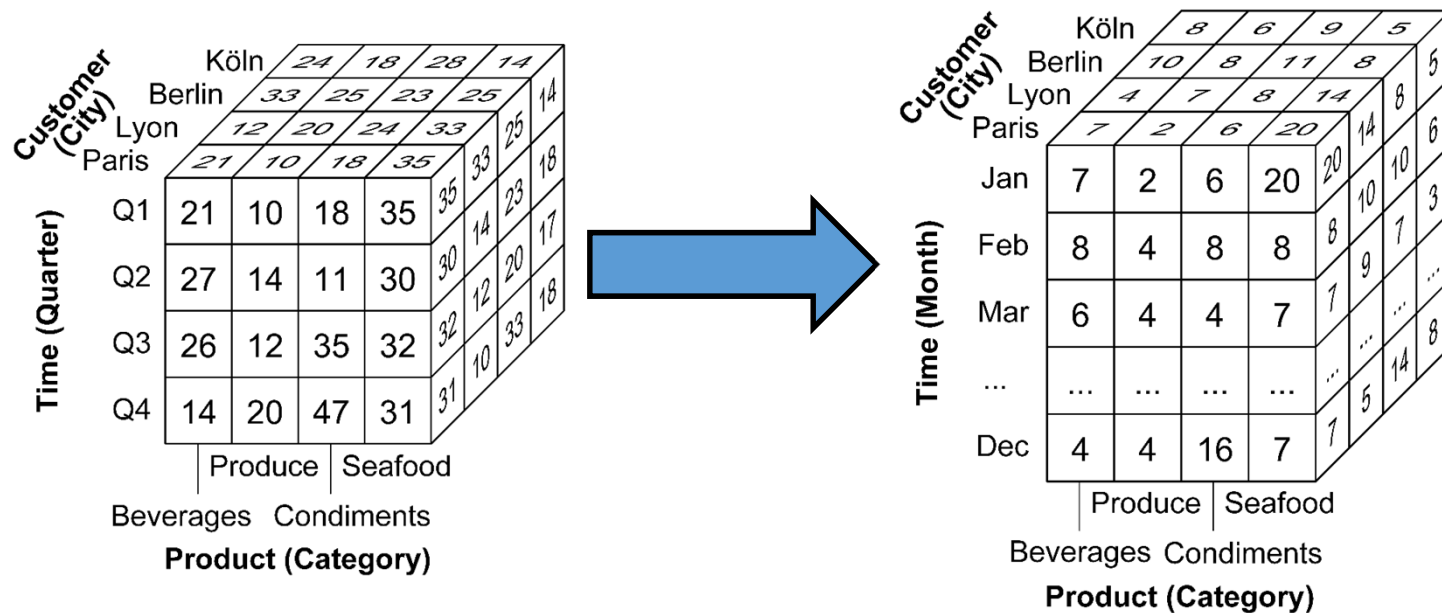
Roll-up (I)



Roll-up (II)



Drill-down



Drill-across

Customer (City)	Time (Quarter)	Product (Category)			
		Condiments	Seafood	Beverages	Produce
Köln	Q1	21	18	10	35
	Q2	27	11	14	30
	Q3	26	35	12	32
	Q4	14	47	20	31
Berlin	Q1	33	23	25	25
	Q2	30	14	20	17
	Q3	32	12	10	33
	Q4	31	10	33	18
Lyon	Q1	12	24	20	33
	Q2	30	14	20	17
	Q3	32	12	10	33
	Q4	31	10	33	18
Paris	Q1	21	18	10	35
	Q2	27	11	14	30
	Q3	26	35	12	32
	Q4	14	47	20	31



Customer (City)	Time (Quarter)	Product (Category)			
		Produce	Seafood	Beverages	Condiments
Köln	Q1	19	12	31	28
	Q2	30	12	10	29
	Q3	28	11	31	28
	Q4	12	22	45	29
Berlin	Q1	30	22	21	26
	Q2	29	14	20	19
	Q3	28	12	31	28
	Q4	29	12	31	28
Lyon	Q1	14	18	22	28
	Q2	29	14	20	19
	Q3	28	12	31	28
	Q4	29	12	31	28
Paris	Q1	19	12	31	28
	Q2	30	12	10	29
	Q3	28	11	31	28
	Q4	12	22	45	29

Customer (City)	Time (Quarter)	Product (Category)			
		Produce	Seafood	Beverages	Condiments
Köln	Q1	19	12	31	28
	Q2	30	12	10	29
	Q3	28	11	31	28
	Q4	12	22	45	29
Berlin	Q1	30	22	21	26
	Q2	29	14	20	19
	Q3	28	12	31	28
	Q4	29	12	31	28
Lyon	Q1	14	18	22	28
	Q2	29	14	20	19
	Q3	28	12	31	28
	Q4	29	12	31	28
Paris	Q1	19	12	31	28
	Q2	30	12	10	29
	Q3	28	11	31	28
	Q4	12	22	45	29

Union

Time (Quarter)	Customer (City)	Product (Category)			
		Produce	Seafood	Beverages	Condiments
Q1	Köln	24	18	28	14
	Berlin	33	25	23	25
	Lyon	12	20	24	33
	Paris	21	10	18	35
Q2		27	14	11	30
Q3		26	12	35	32
Q4		14	20	47	31

Time (Quarter)	Customer (City)	Product (Category)			
		Produce	Seafood	Beverages	Condiments
Q1	Bilbao	24	18	28	14
	Madrid	22	27	23	15
		33	22	18	23
		14	13	25	17
Q2		20	18	23	17
Q3					
Q4					



Time (Quarter)	Customer (City)	Product (Category)			
		Produce	Seafood	Beverages	Condiments
Q1	Madrid	22	27	23	15
	Bilbao	24	18	28	14
	Köln	24	18	28	14
	Berlin	33	25	23	25
Q2	Lyon	12	20	24	33
	Paris	21	10	18	35
		27	14	11	30
		26	12	35	32
Q3		14	20	47	31
Q4					

A. Vaisman & E.Zimanyi

Difference

Customer (City)	Köln				Berlin				Lyon				Paris						
	24	18	28	14	33	25	23	25	12	20	24	33	21	10	18	35			
	21	10	18	35	35	14	23	17	27	14	11	30	26	12	35	32			
	30	12	20	18	32	10	33	18	14	20	47	31	14	20	47	31			
	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4			
Produce				Seafood				Beverages				Condiments				Product (Category)			

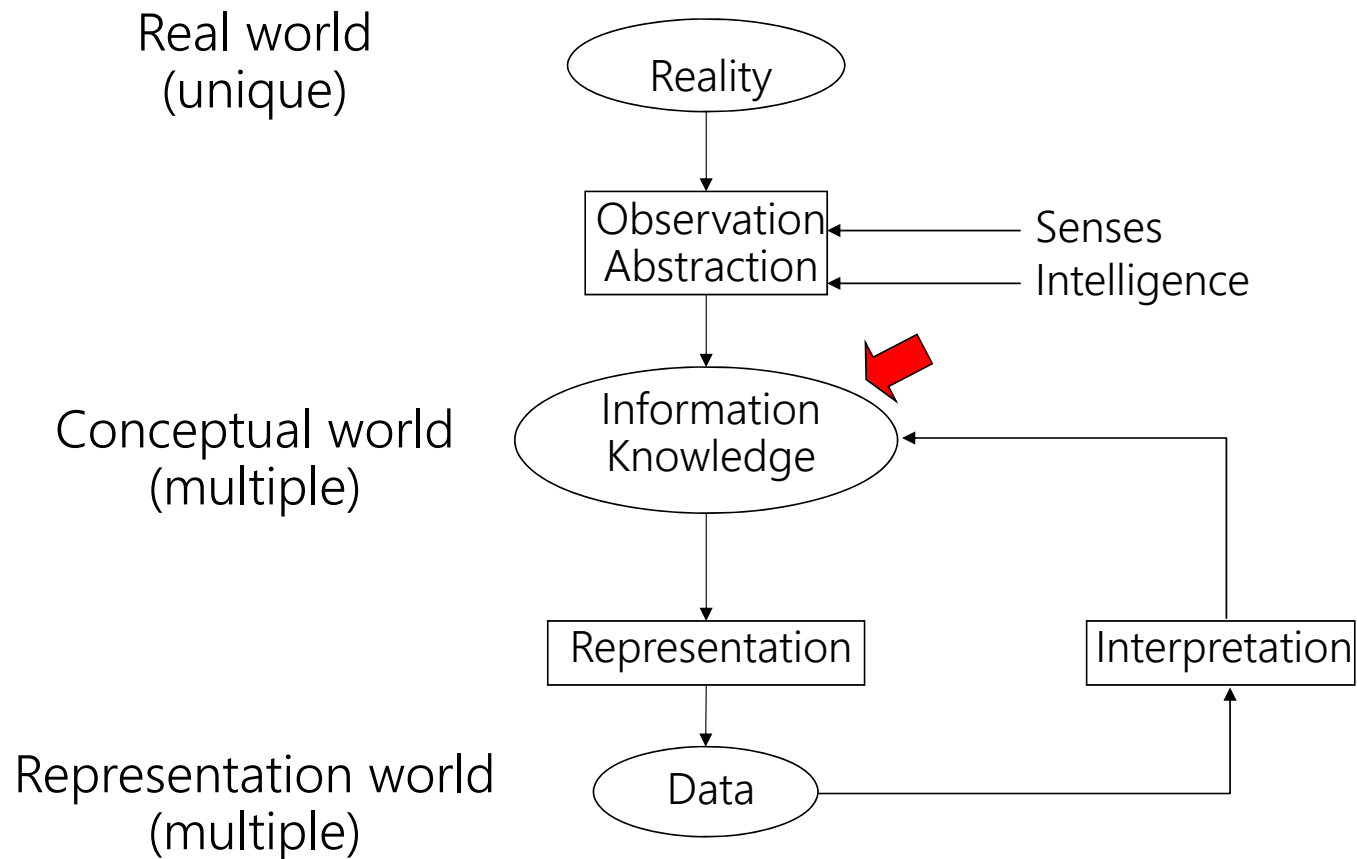
Customer (City)	Köln				Berlin				Lyon				Paris				
	24				28				25				25				
	33				25				33				25				
	20				33				35				33				
	21				35				33				25				
Time (Quarter)	Q1	21				35				35				23			
	Q2	27				30				30				20			
	Q3					35				32				33			
	Q4					47				31				31			
		Produce				Seafood				Beverages				Condiments			
Product (Category)																	



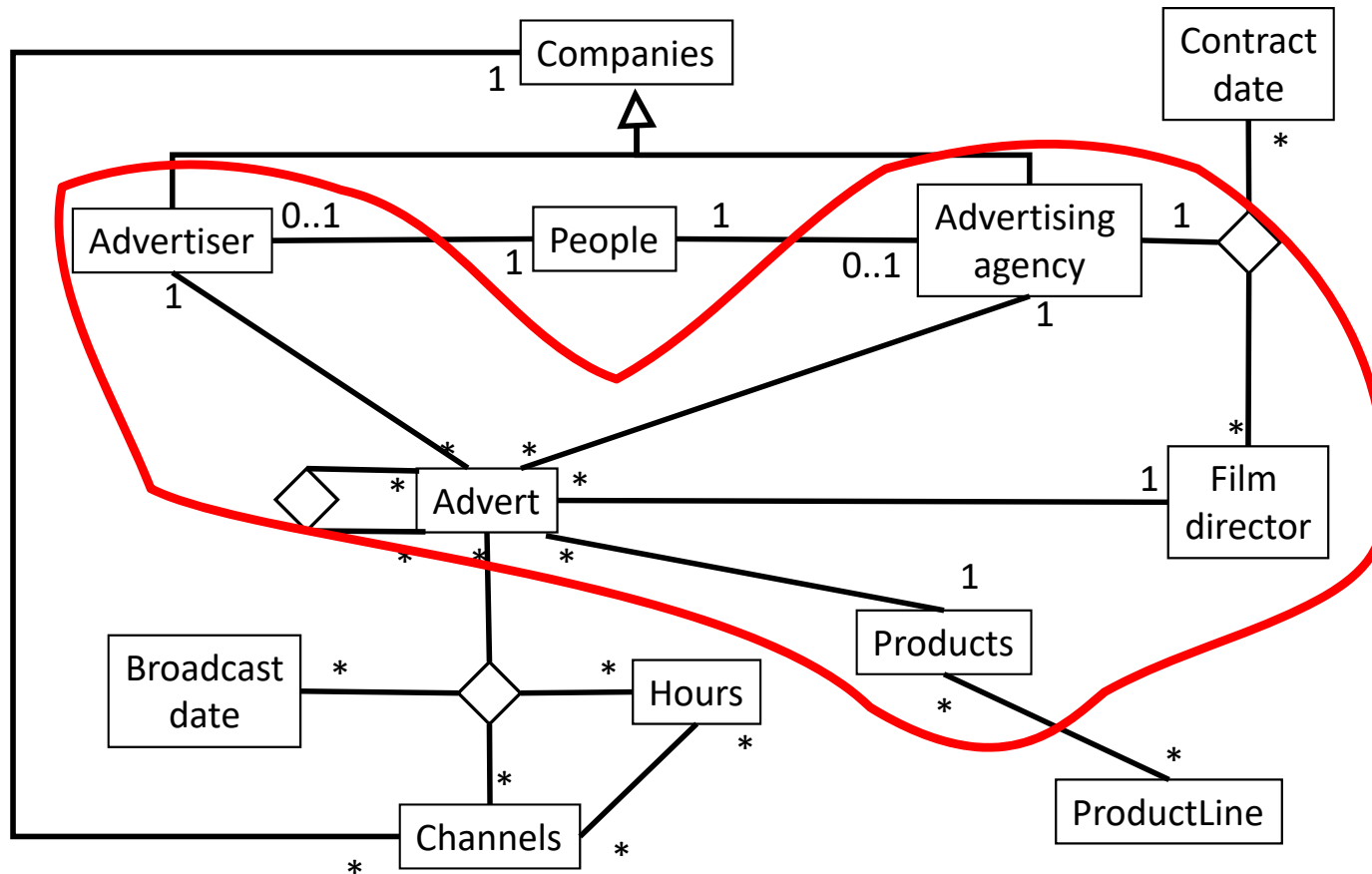
Customer (City)	Köln				Berlin				Lyon				Paris			
Time (Quarter)	Q1		10	18												
	Q2		14	11												
	Q3	26	12													
	Q4	14	20													
Produce				Seafood				Beverages				Condiments				
Product (Category)																

Conceptual world

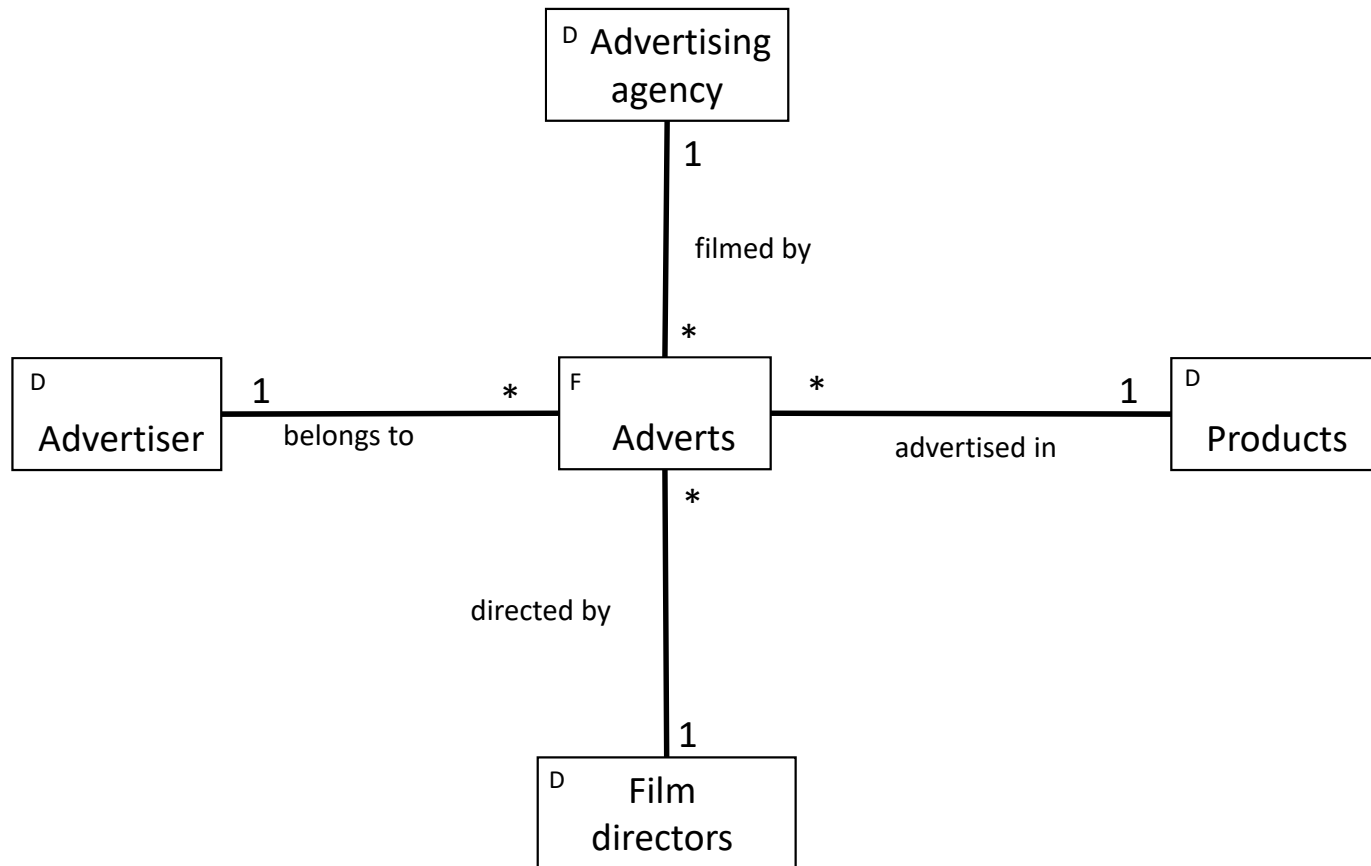
Three worlds



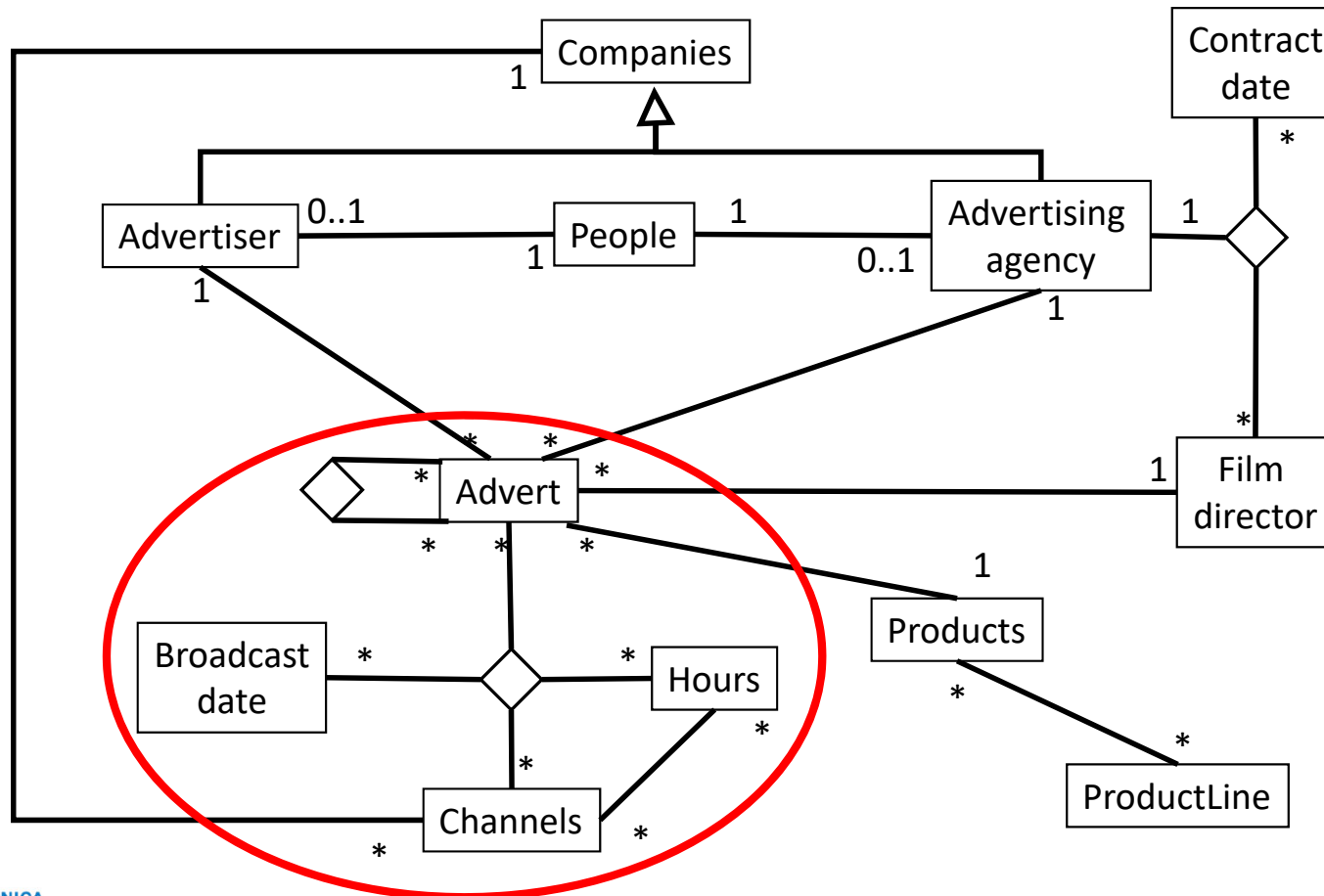
Example of transactional modeling (UML)



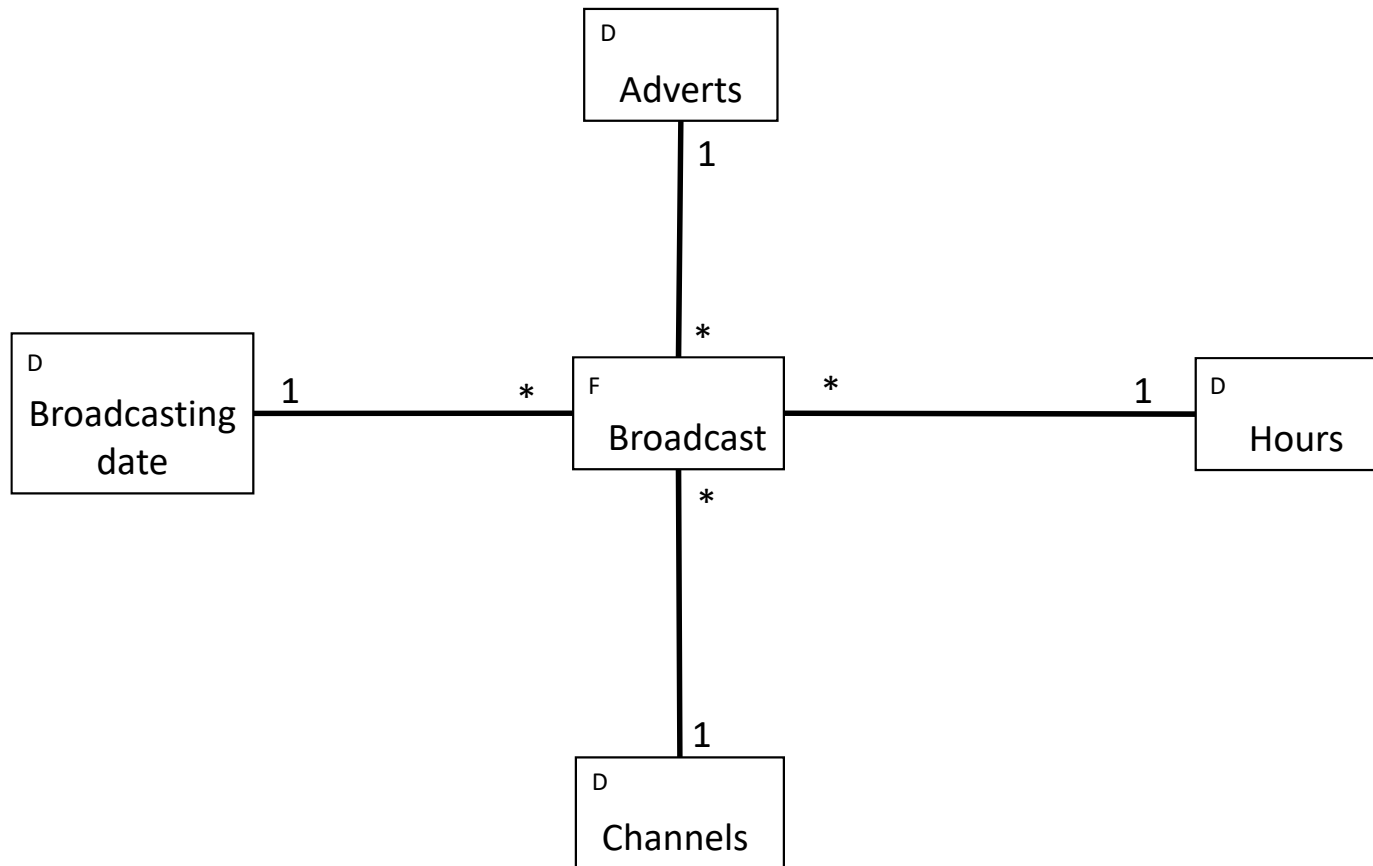
Star schema (I)



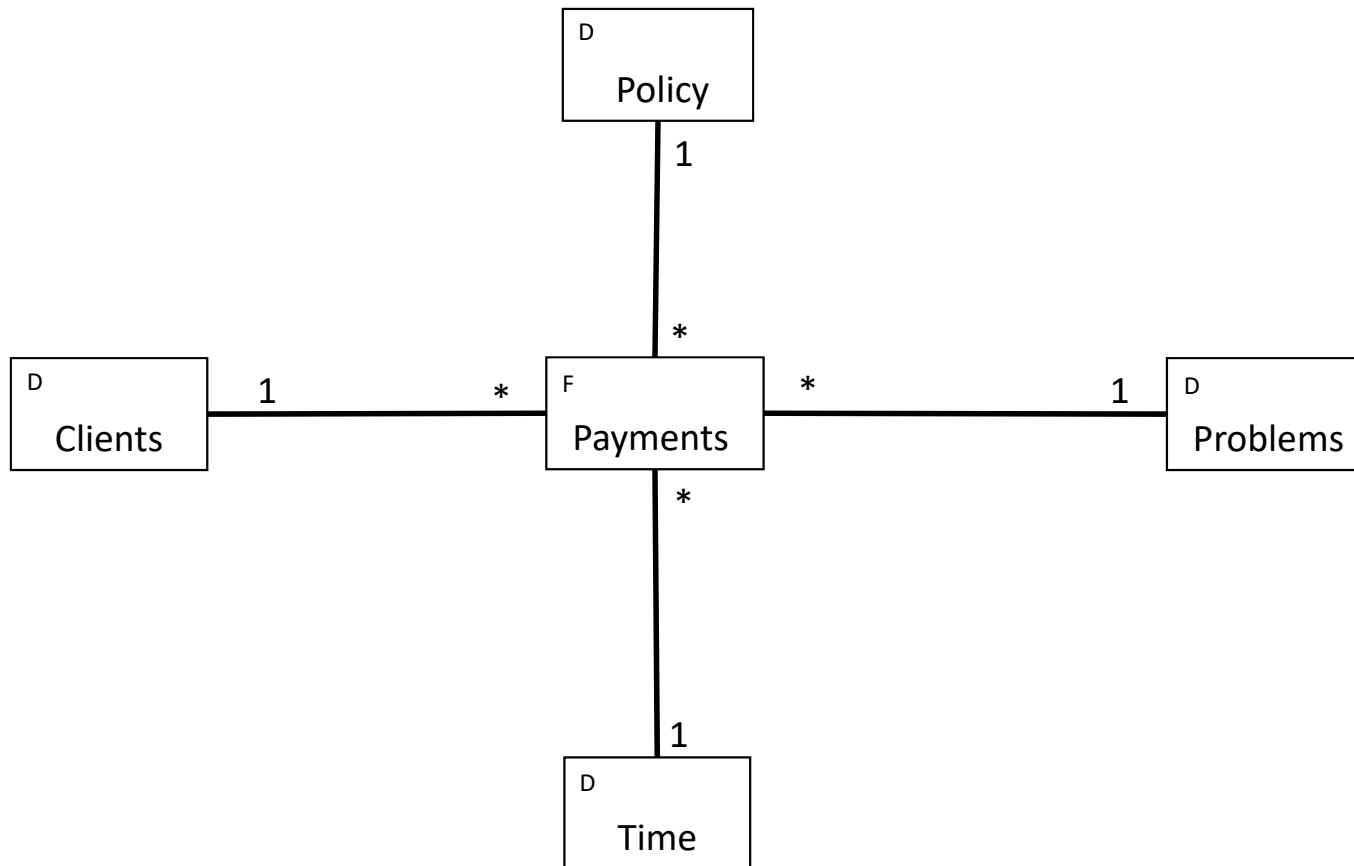
Example of transactional modeling (UML)



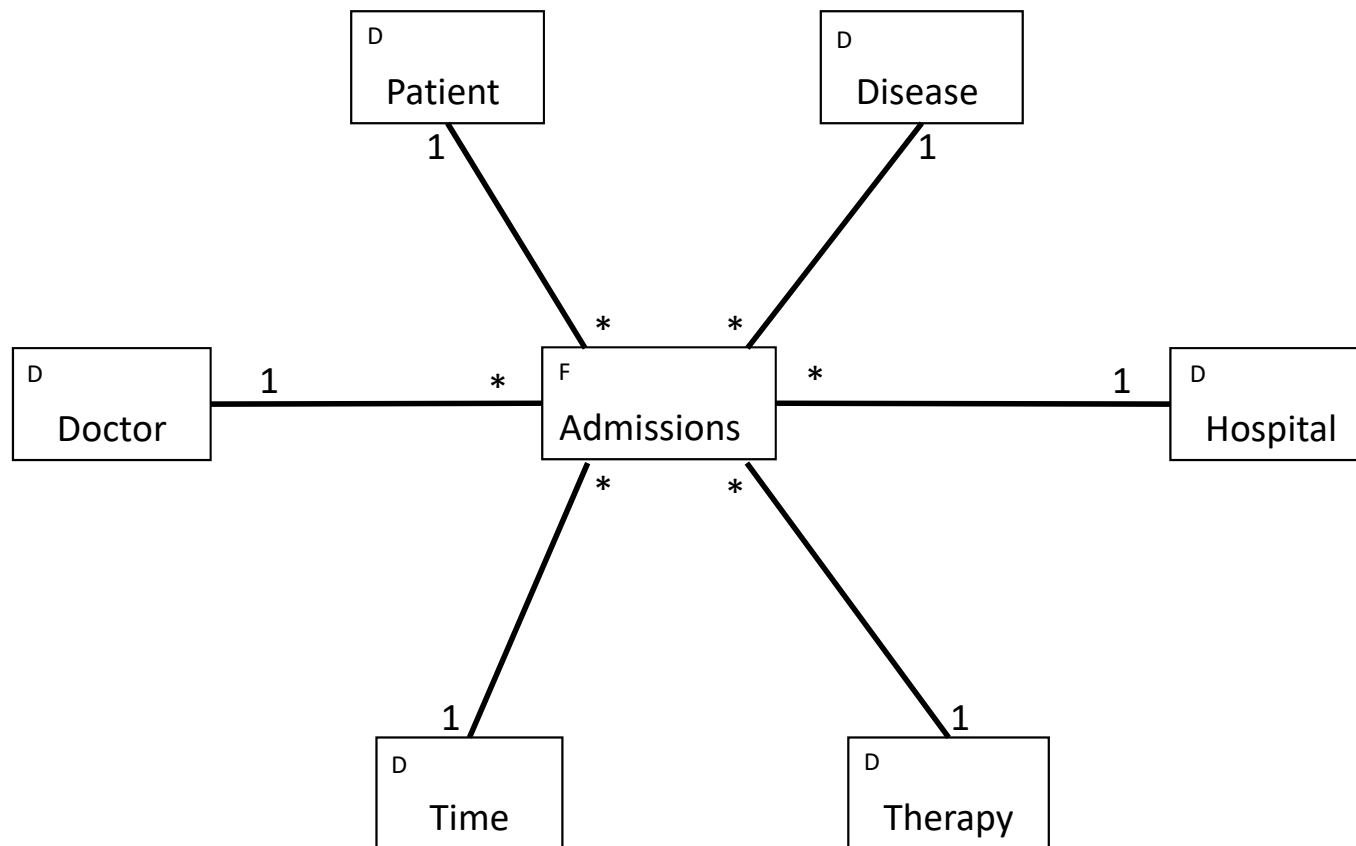
Star schema (II)



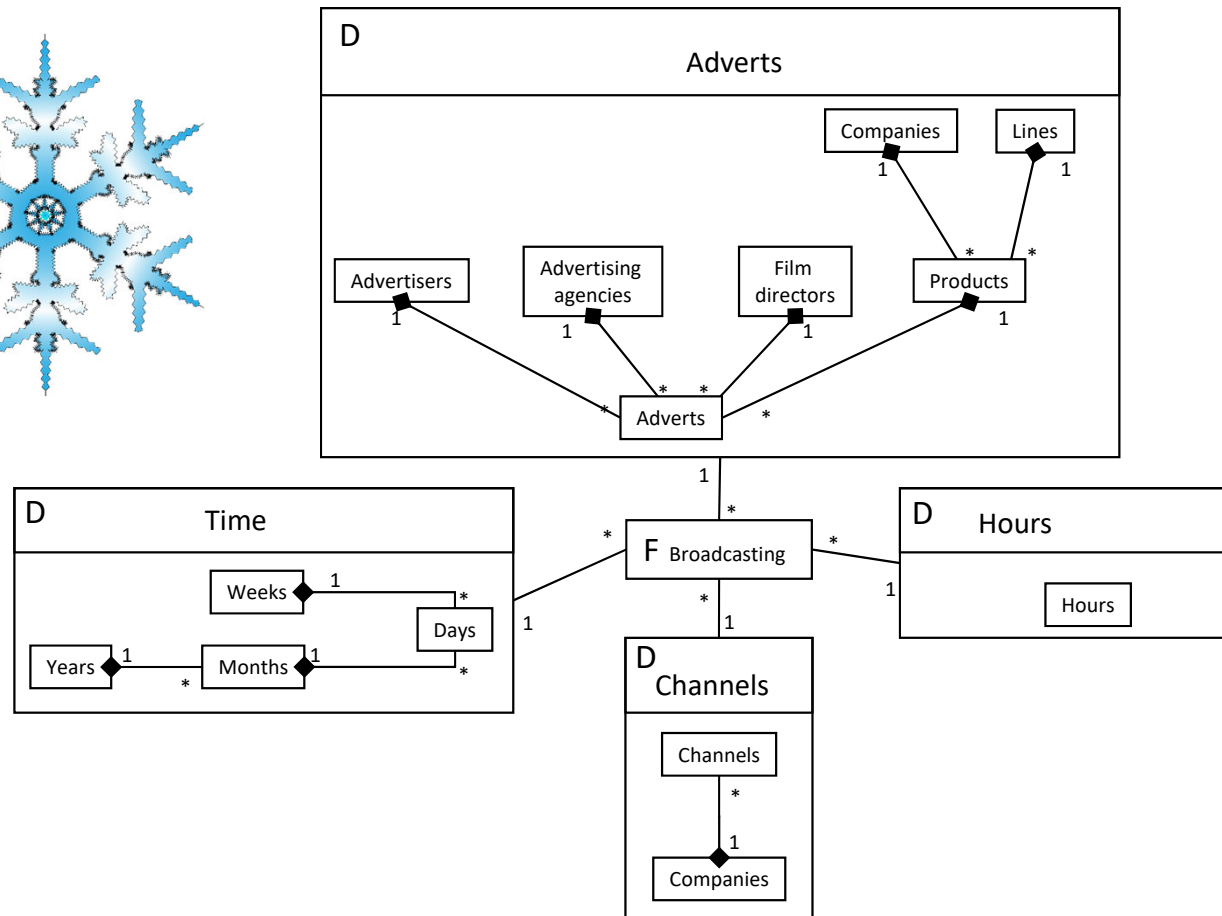
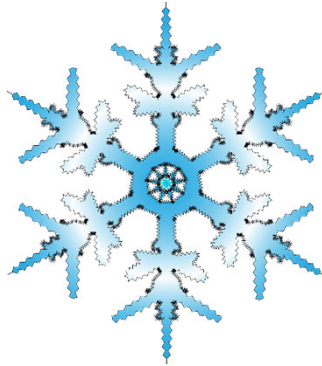
Star schema (III)



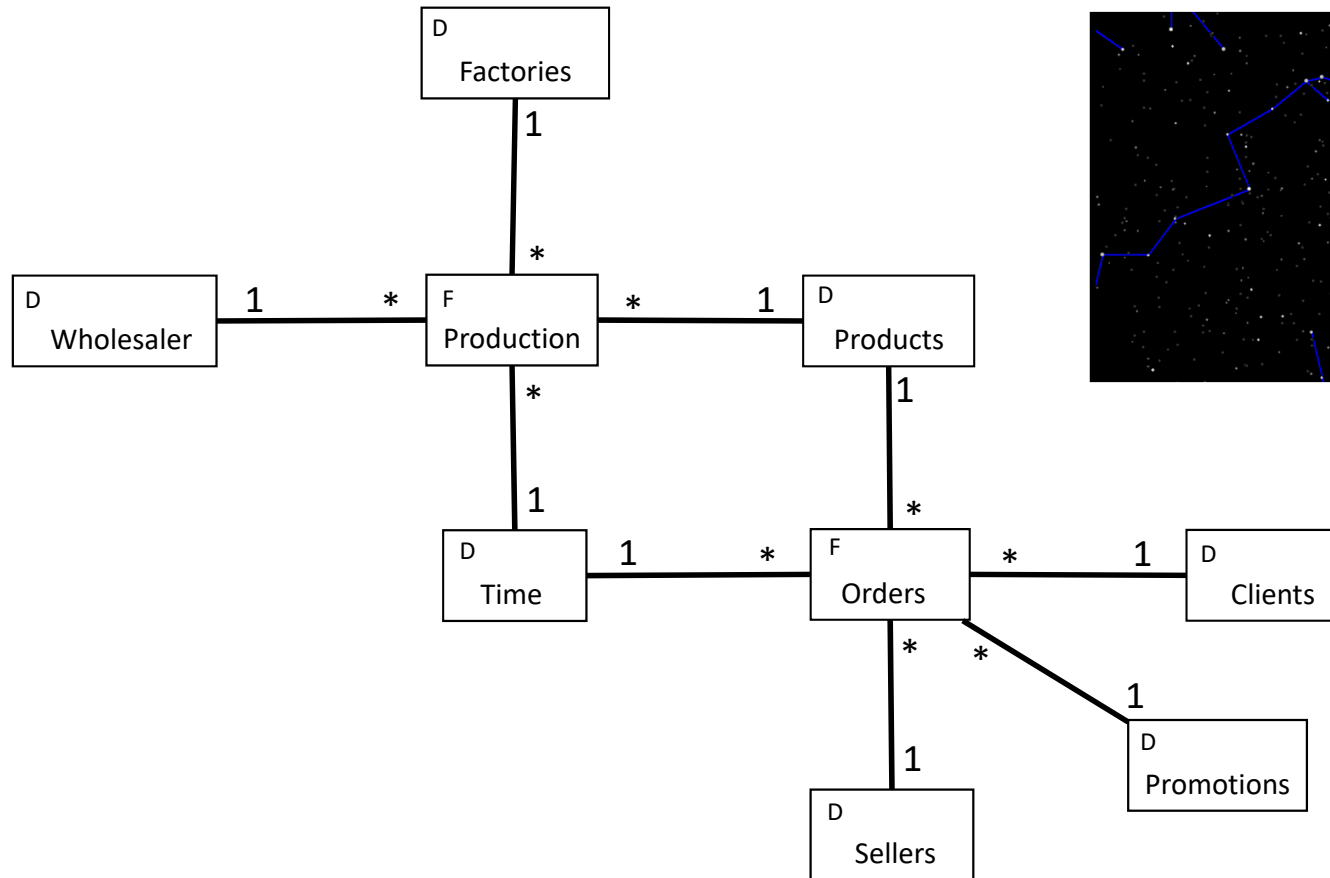
Star schema (IV)



Snowflake

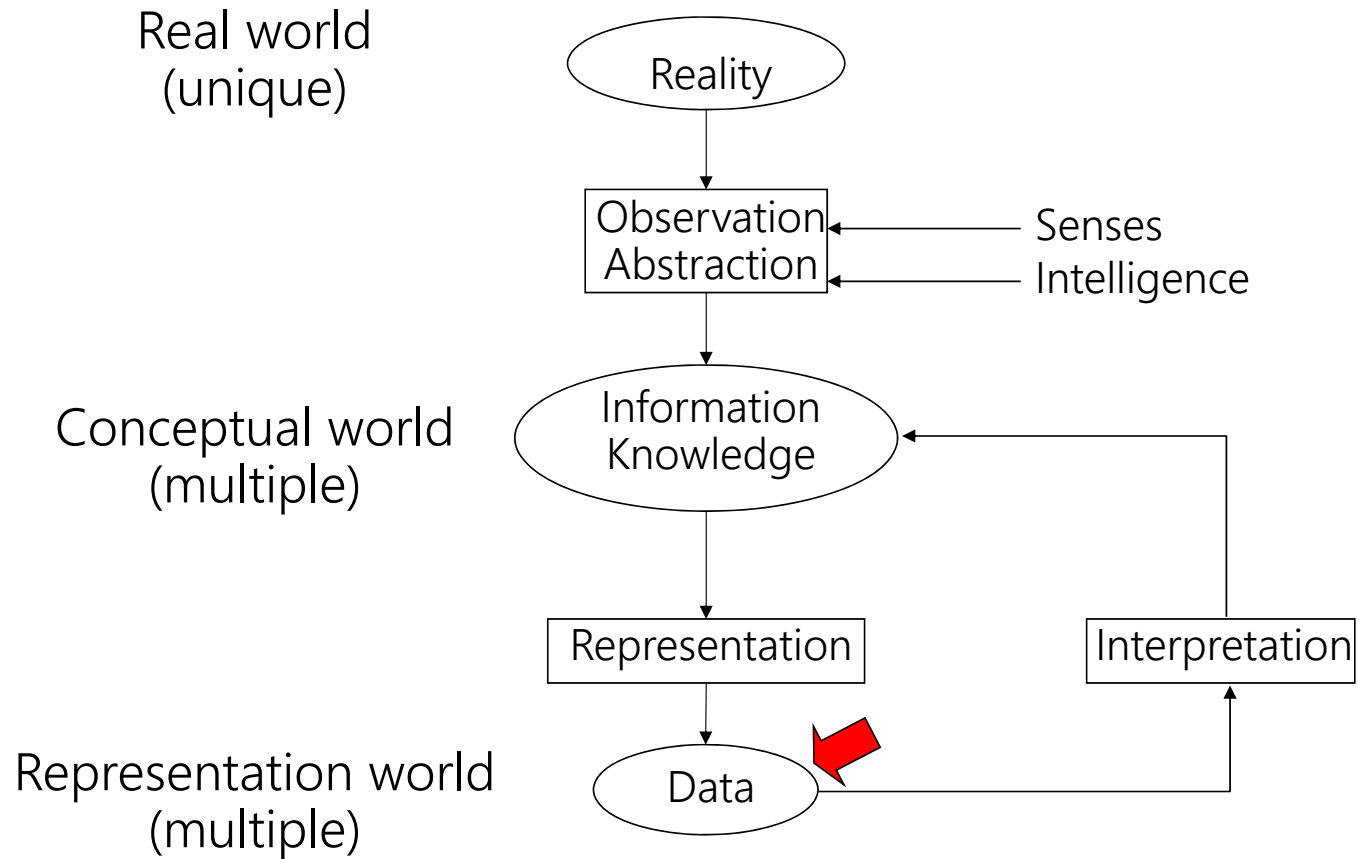


Galaxy or Constellation

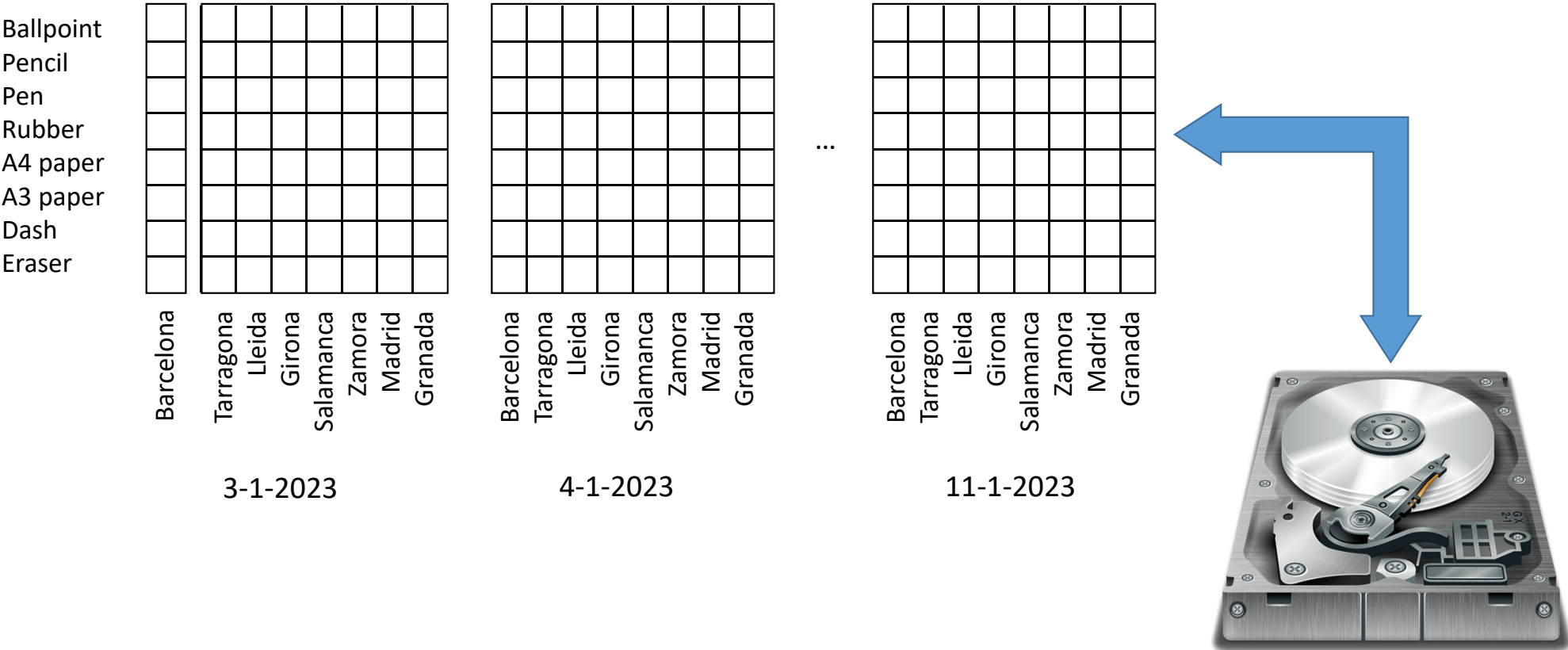


Representations world

Three worlds

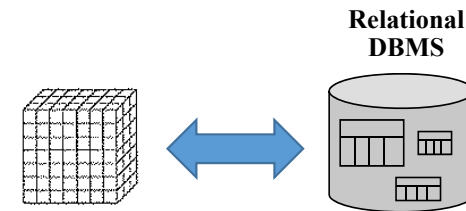


MOLAP: Multidimensional matrix

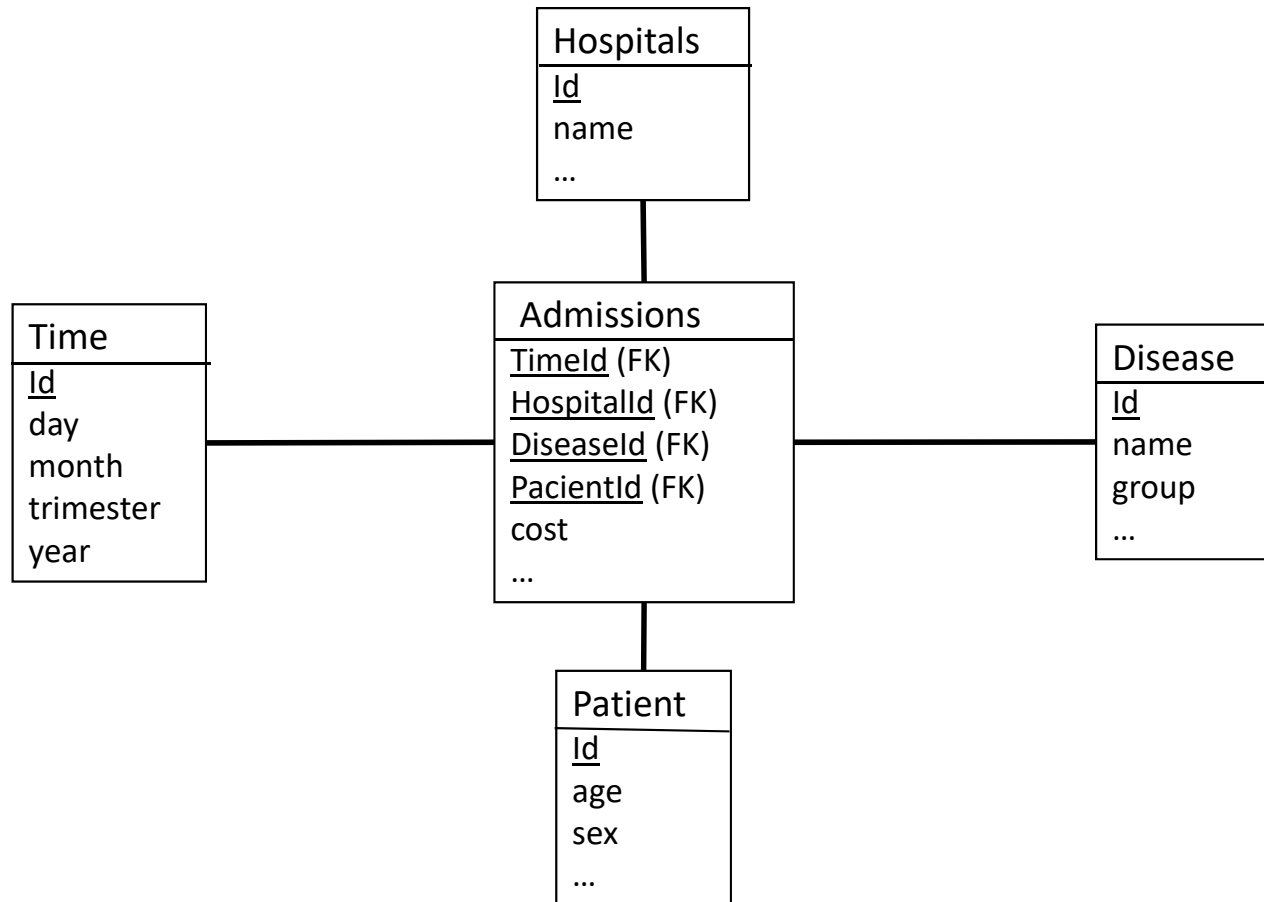


ROLAP: Characteristics

- Relational DBMS with multidimensional views
 - Two levels: Storage and Translation
- Use standard SQL
 - Easy to obtain
 - Independent of the DBMS
- Performance issues
 - Relational DBMS conceived for OLTP (later improved/adapted)
 - Conceptual OLAP operations are missing
 - Generates too many joins
 - Uses specific techniques



ROLAP: Star-join schema



ROLAP: Cube-Query

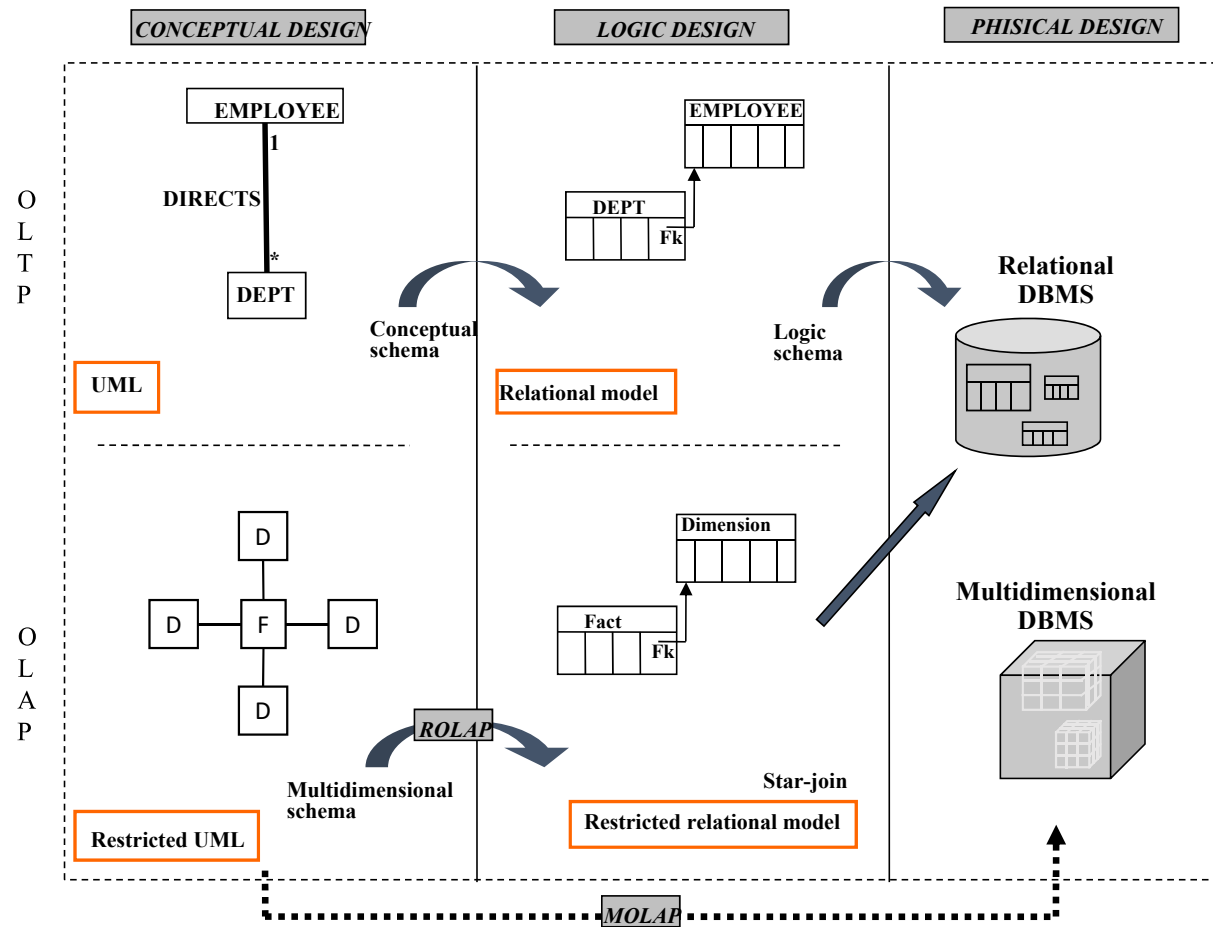
```
SELECT d1.attr, ..., dn.attr, F(f.Measure1), ...  
FROM Fact f, Dimension1 d1, ..., Dimensionn dn  
WHERE f.key1 = d1.ID AND ... AND f.keyn = dn.ID AND <slice-dice condition>  
GROUP BY d1.attr, ..., dn.attr  
ORDER BY d1.attr, ..., dn.attr
```

```
SELECT h.name AS Hospital , t.month AS Month, AVG(cost)  
FROM Admissions a, Hospitals h, Time t  
WHERE a.HospitalId=h.Id AND a.TimeId=t.id AND t.year=2022  
GROUP BY h.name, t.month  
ORDER BY h.name, t.month
```

ROLAP: Results table

Hospital	Month	Average Cost
Duran i Reinalts	January'22	3300
Duran i Reinalts	February'22	4500
Duran i Reinalts	...	...
Duran i Reinalts	All	4300
Bellvitge	January'22	180
Bellvitge	February'22	300
Bellvitge	...	...
Bellvitge	All	200

Comparison of design steps



HOLAP

Hybrid approach:

- a) Store dense chunks in MOLAP and sparse ones in ROLAP
- b) Store atomic data in ROLAP and aggregates in MOLAP
- c) Store frequently accessed data in MOLAP and the remaining in ROLAP

Closing

Summary

1. OLAP definition
2. Cube
3. Multidimensional schemas
 - Star
 - Snowflake
 - Galaxy or Constellation
4. Kinds of multidimensional tools
 - ROLAP
 - MOLAP
 - HOLAP
5. Star-join schema
6. Cube-Query

References

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- BI-verdict. www.bi-verdict.com, June 2010
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